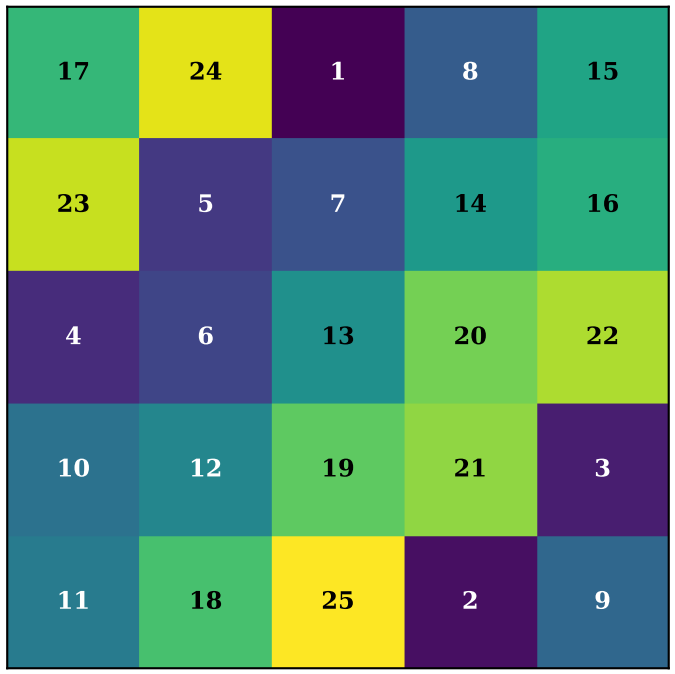
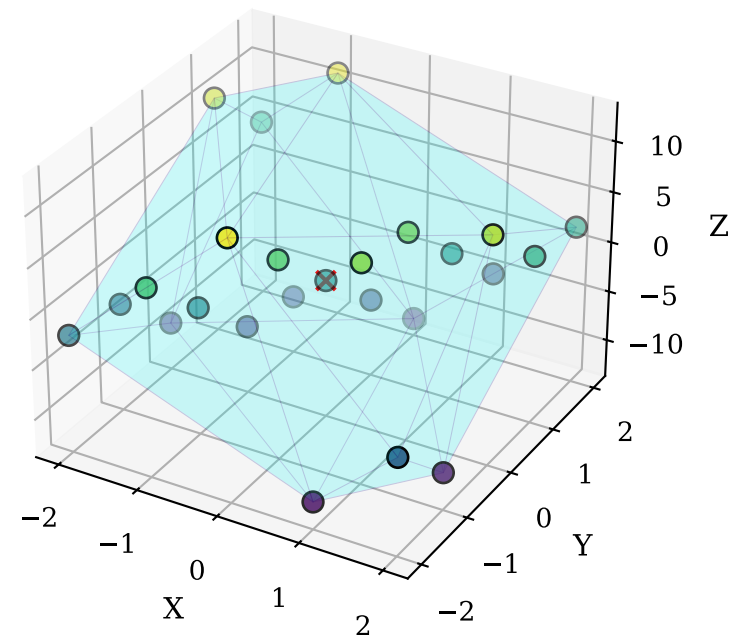
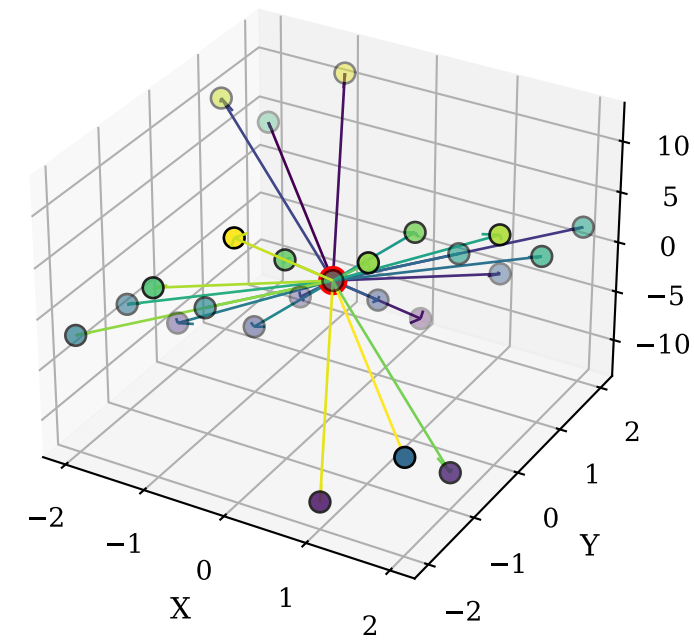
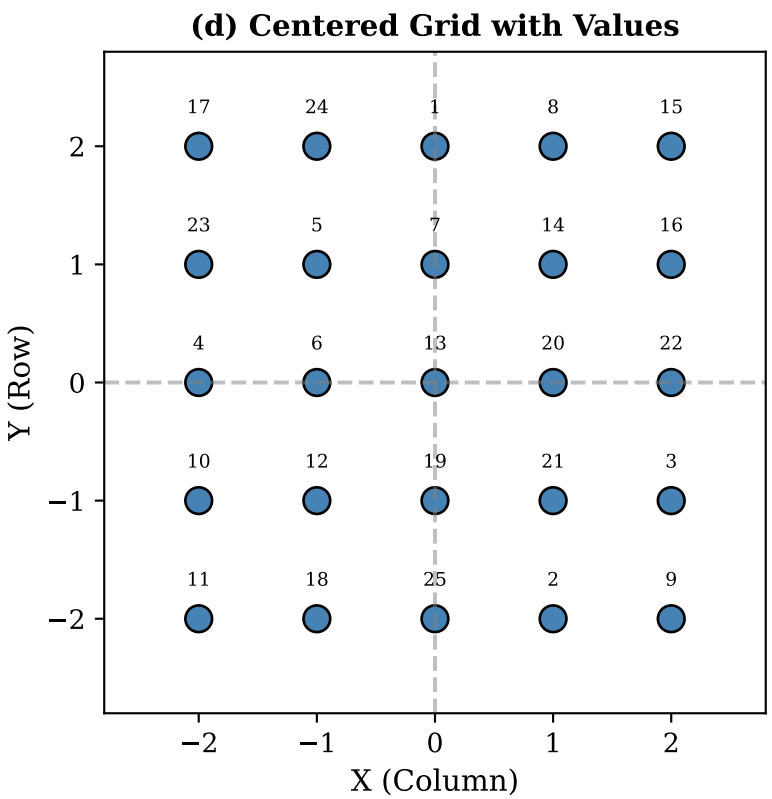
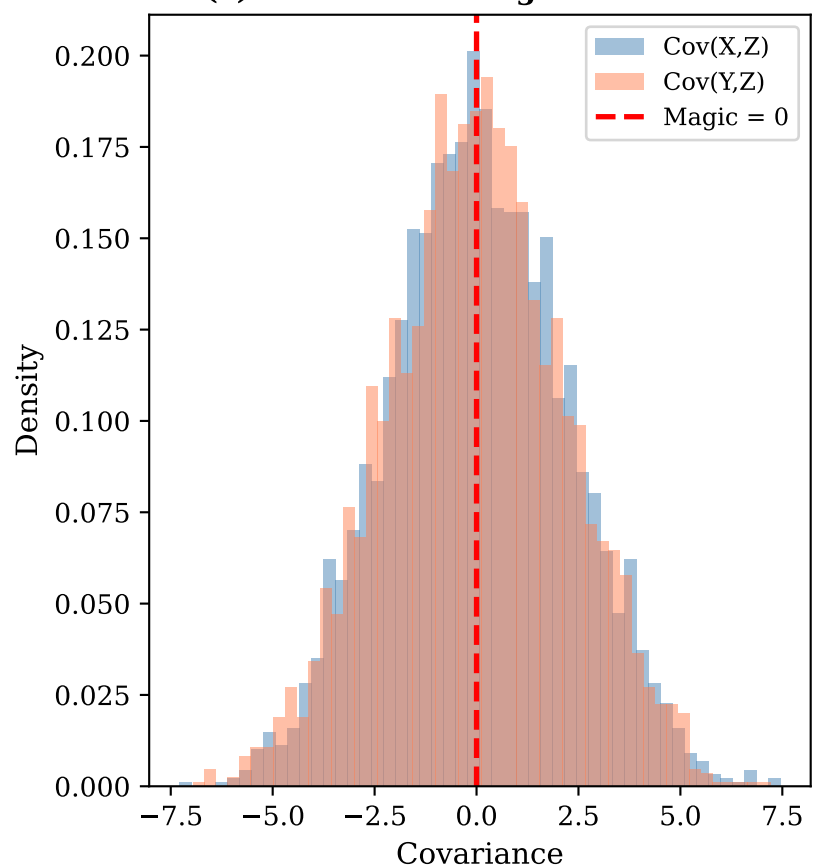
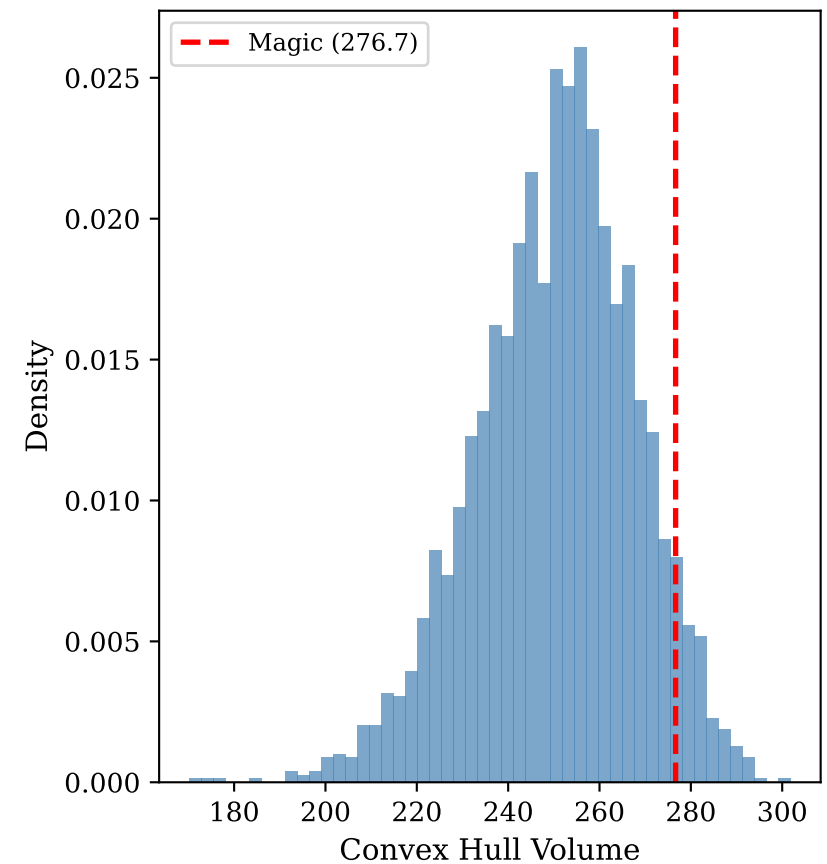

(a) 5×5 Magic Square
M(5) = 65
17 24 1 8 15
23 5 7 14 16
4 6 13 20 22
10 12 19 21 3
11 18 25 2 9


(b) 5×5 Magic Gem
X
Y
Z


(c) 25 Vectors from Centroid
X
Y


(d) Centered Grid with Values
17 24 1 8 15
23 5 7 14 16
4 6 13 20 22
10 12 19 21 3
11 18 25 2 9
X (Column)
Y (Row)


(e) Covariance: Magic vs Random
Cov(X,Z)
Cov(Y,Z)
Magic = 0
Density
Covariance


(f) Hull Volume Distribution
Magic (276.7)
Density
Convex Hull Volume